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0000 separate (IDL.SEM_PHASE)¶
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; -- GET_BASE_STRUCT¶
¶ function IS_STATIC_RANGE (A: TREE) return Boolean is¶ begin¶ return A.TY = DN_RANGE and then GET_STATIC_VALUE (D (AS_EXP1, A)) /= TREE_VOID and
then GET_STATIC_VALUE (D (AS_EXP2, A)) /= TREE_VOID;¶ end IS_STATIC_RANGE;¶ function IS_STATIC_SUBTYPE (A : TREE) return Boolean is¶ begin¶ i
f A.TY in CLASS_PRIVATE_SPEC then¶ return GET_BASE_STRUCT (A).TY in CLASS_SCALAR and then IS_STATIC_SUBTYPE (D (SM_TYPE_SPEC, A));¶ elsif A.TY
= DN_INCOMPLETE then¶ return GET_BASE_STRUCT (A).TY in CLASS_SCALAR and then IS_STATIC_SUBTYPE (D (XD_FULL_TYPE_SPEC, A));¶ end if;¶ if A
.TY not in CLASS_SCALAR then¶ return False;¶ end if;¶ if D (SM_BASE_TYPE, A) = A then¶ -- THIS IS A SCALAR BASE TYPE; TEST-FOR GENERI
C FORMAL TYPE¶ return D (SM_RANGE, A) /= TREE_VOID;¶ else¶ -- THIS A SUBTYPE, NOT A BASE TYPE¶ return IS_STATIC_RANGE (D (SM_RANGE,
A)) and then IS_STATIC_SUBTYPE (D (SM_TYPE_SPEC, D (SM_RANGE, A)));¶ end if;¶ end IS_STATIC_SUBTYPE;¶ function IS_STATIC_DISCRETE_RANGE (A : TRE
0010 E) return Boolean is¶ begin¶ case A.TY is¶ when DN_DISCRETE_SUBTYPE =>¶ return IS_STATIC_DISCRETE_RANGE (D (AS_NAME, A));¶ else
¶ return IS_STATIC_DISCRETE_RANGE (D (AS_NAME, A)) and then IS_STATIC_RANGE (D (AS_CONSTRAINT, A));¶ end if;¶ when DN_RANGE=>¶ ret
urn IS_STATIC_RANGE (A);¶ when DN_USED_NAME_ID =>¶ return D (SM_DEFN, A) /= TREE_VOID and then IS_STATIC_DISCRETE_RANGE (D (SM_EXP_TYPE, D
(SM_DEFN, A)));¶ when DN_SELECTED =>¶ return IS_STATIC_DISCRETE_RANGE (D (AS_DESIGNATOR, A));¶ when others =>¶ return False;¶
end case;¶ end IS_STATIC_DISCRETE_RANGE;¶ function IS_STATIC_INDEX_CONSTRAINT (ARRAY_TYPE, INDEX_CONSTRAINT : TREE) return Boolean is¶ INDEX
_LIST: SEQ_TYPE := LIST (D (AS_DISCRETE_RANGE_S, INDEX_CONSTRAINT));¶ INDEX: TREE;¶ ARRAY_TYPE_SPEC: TREE := ARRAY_TYPE;¶ begin¶ while
not IS_EMPTY (INDEX_LIST) loop¶ POP (INDEX_LIST, INDEX);¶ if not IS_STATIC_DISCRETE_RANGE (INDEX) then¶ return False;¶ end if;¶
end loop;¶ if ARRAY_TYPE_SPEC.TY in CLASS_PRIVATE_SPEC then¶ ARRAY_TYPE_SPEC := D (SM_TYPE_SPEC, ARRAY_TYPE_SPEC);¶ elsif ARRAY_TYPE_S
PEC.TY = DN_INCOMPLETE then¶ ARRAY_TYPE_SPEC := D (XD_FULL_TYPE_SPEC, ARRAY_TYPE_SPEC);¶ end if;¶ if ARRAY_TYPE_SPEC.TY = DN_CONSTRAINED_A
0020 RRAY then¶ -- $$$ SHOULD NOT HAPPEN [RM 3.6.1/3]¶ return False;¶ end if;¶ INDEX_LIST := LIST (D (SM_INDEX_S, ARRAY_TYPE_SPEC));¶
while not IS_EMPTY (INDEX_LIST) loop¶ POP (INDEX_LIST, INDEX);¶ if not IS_STATIC_SUBTYPE (D (SM_TYPE_SPEC, INDEX)) then¶ return False
;¶ end if;¶ end loop;¶ return True;¶ end IS_STATIC_INDEX_CONSTRAINT;¶
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end CHK_STAT;

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